

FIT5196-S2-2026 Group Assessment

Stage 1 (15%)

This is a group assessment and worth 15% of your total mark.

Due date: 23:55, Wednesday, Week 7 (9 September 2026)

This specification defines the assessed work, shared transformation rules, submission requirements, quick-start guidance, checklist and glossary. The separate marking rubric defines how evidence is evaluated. The [public_data_dictionary.csv](#) in your allocated data package defines the required output table grains, fields, field order, data types, nullability and comparison rules. The package [README.md](#) defines package-specific source conventions.

1. Scenario

Your team has received two historical exports from different operational systems used by a Melbourne technology retailer. One export is JSON and the other is XML. Together they describe customers, products, transactions, deliveries and long customer reviews.

The systems do not represent the same information in the same way. They use different nesting, field names, data formats and text conventions. Records may overlap within and across the two sources. Your team must build a reproducible workflow that parses the sources, reconciles overlapping records, produces six analysis-ready relational tables, validates the result and conducts a focused exploratory data analysis (EDA).

This is the first stage of the FIT5196 assessment project. Later assessment stages will use related current-period data. The teaching team will provide certified data for high-stakes downstream assessment so that an A1 mistake is not automatically charged again later. You should still retain your code, decisions and outputs because your reasoning may inform later work.

2. Learning outcomes assessed

By completing A1, your group should demonstrate that it can:

1. parse and inspect semi-structured JSON and XML using appropriate structured parsers;
2. map nested and repeated source records to tables with defined grains and relationships;
3. standardise identifiers, dates, timestamps, booleans, numeric values and narrative fields;
4. use regular expressions for bounded text cleaning and extraction tasks;
5. preserve valid multilingual Unicode text while creating a separate purpose-specific Latin-script analysis field;
6. reconcile duplicate and overlapping records without double counting;

FIT5196-S2-2026 小组评估第一阶段（15%）

这是一个小组评估，占你总成绩的15%。

截止日期：23:55，星期三，第7周（2026年9月9日）

本规范定义了评估工作、共享转换规则、提交要求、快速入门指南、检查清单和术语表。单独的评分标准定义了如何评估证据。您分配的数据包中的 `public_data_dictionary.csv` 定义了所需的输出表粒度、字段、字段顺序、数据类型、可空性和比较规则。包内的 `README.md` 定义了包特定的源规范。

1. 场景

您的团队收到了来自墨尔本一家科技零售商使用的两个不同操作系统的历史导出文件。一个是 JSON 格式，另一个是 XML 格式。它们共同描述了客户、产品、交易、配送和长期客户评论。

这些系统以不同的方式表示相同的信息。它们使用不同的嵌套方式、字段名、数据格式和文本规范。记录可能在两个来源内部或跨来源重叠。您的团队必须构建一个可重复的工作流程，该工作流程解析来源、协调重叠的记录、生成六个可用于分析的关系表、验证结果并进行有针对性的探索性数据分析（EDA）。

这是FIT5196评估项目的第一个阶段。后续的评估阶段将使用相关的当期数据。教学团队将提供经过认证的数据，用于高风险的下游评估，以便A1错误不会在后期自动再次被扣分。您仍然应该保留您的代码、决策和输出，因为您的推理可能会为后续工作提供参考。

2. 评估的学习成果

通过完成A1，你的小组应该展示出它能够：

1. 使用适当的结构化解析器解析和检查半结构化JSON和XML；
2. 将嵌套和重复的来源记录映射到具有定义粒度和关系的关系表；
3. 标准化标识符、日期、时间戳、布尔值、数值和叙述性字段；
4. 使用正则表达式进行有界文本清理和提取任务；
5. 在创建特定用途的分析字段时，保留有效的多语言 Unicode 文本拉丁字母脚本；
6. 调整重复和重叠的记录，避免重复计数；

- 7. validate keys, relationships, arithmetic and transformation behaviour;
- 8. conduct defensible univariate, bivariate and multivariate EDA; and
- 9. translate EDA evidence into feasible machine-learning questions while recognising leakage, evaluation and deployment risks.

Model training is not required in A1.

3. Files supplied

Each group receives one allocated archive named `GroupNNN_A1.zip`, where `NNN` is the three-digit group number. It contains:

- `A1_manifest.json`
- `README.md`
- `public_data_dictionary.csv`
- `raw_input/`
 - `GroupNNN_commerce.json`
 - `GroupNNN_operations.xml`

The teaching team also supplies the marking rubric and one `templates/` folder:

- `A1_source_to_target_mapping_template.csv`;
- `A1_solution_template.ipynb`;
- `A1_EDA_template.ipynb`;
- `A1_public_text_test_cases.csv`; and
- `A1_AI_use_declaration_template.docx`.

The source files contain hundreds of customer and product records and thousands of transaction, delivery and review records. These are scale indicators, not expected clean row counts. The teaching team does not publish the canonical row counts, overlap counts, duplicate locations or expected output values.

Use only the data allocated to your group. Some source content may be shared across groups, but each package is assessed against its own certified reference. Before beginning, verify the group number in the archive name, source filenames and public manifest.

4. Required work

Task 1: Parse and profile the sources

Use a JSON parser for the JSON file and an XML parser for the XML file. Do not parse either structured document as plain text with regular expressions.

- 7. 验证键、关系、算术和转换行为；
- 8. 进行可靠的单元变量、双变量和多变量EDA；以及
- 9. 将EDA证据转化为可行的机器学习问题，同时识别泄露、评估和部署风险。

在A1中不需要进行模型训练。

3. 提供的文件

每个小组将获得一个分配的存档，命名为`GroupNNN_A1.zip`，其中`NNN`是三位数的小组编号。它包含：

- `A1_manifest.json`
- `README.md`
- `public_data_dictionary.csv`
- `raw_input/`
 - `GroupNNN_commerce.json`
 - `GroupNNN_operations.xml`

教学团队还提供评分标准和一个 `templates/` 文件夹：

- 一个`1_源_到_目标_映射_模板.csv`；
- 一个`1_解决方案_模板.ipynb`；
- 一个`1_EDA_模板.ipynb`；
- `A1_公共_文本_测试_案例.csv`；和
- 一个`1_AI_使用_声明_模板.docx`。

源文件包含数百个客户和产品记录，以及数千笔交易、交付和评价记录。这些是规模指标，不是预期的干净行数。教学团队不会发布规范行数、重叠计数、重复位置或预期输出值。

仅使用分配给您小组的数据。部分源内容可能跨小组共享，但每个包都根据其自身的认证参考进行评估。开始前，请核实存档名称、源文件名和公共清单中的组号。

4. 必要工作

任务 1：解析和分析源

使用 JSON 解析器处理 JSON 文件，使用 XML 解析器处理 XML 文件。不要用正则表达式将这两种结构化文档当作纯文本解析。

Your notebook and source-to-target mapping must show that you investigated the following. The report should summarise only the material decisions:

- major objects, nested arrays, repeated XML elements and source-specific representations;
- candidate primary keys and foreign keys;
- the grain of each source collection;
- date, timestamp, boolean, currency, percentage and missing-value formats;
- fields that appear in one or both sources;
- duplicate records within a source and overlapping records across sources; and
- assumptions needed before transformation.

Complete the supplied source-to-target mapping template. Its target fields are pre-filled, but you must investigate and enter the applicable JSON and/or XML path, derivation or normalisation, overlap/conflict rule and notebook evidence for every required output field. Source paths are intentionally not pre-filled because the two source layouts differ across allocated packages. The mapping must describe your method, not list every individual data row

Complete the six blank columns using JSON, XML, both or derived for source_format; structural paths for json_source_path and xml_source_path; and concise entries for transformation_or_derivation, overlap_or_conflict_rule and notebook_evidence. Use | to separate multiple input fields. The solution notebook template explains each column and contains fictional worked examples..

Task 2: Produce six standardised relational tables

Produce exactly these six submitted data tables:

1. orders
2. order_items
3. customers
4. deliveries
5. products
6. product_reviews

Use these filenames:

- GroupNNN_orders_standardised.csv
- GroupNNN_order_items_standardised.csv
- GroupNNN_customers_standardised.csv
- GroupNNN_deliveries_standardised.csv
- GroupNNN_products_standardised.csv
- GroupNNN_product_reviews_standardised.csv

The required grains and primary keys are:

你的笔记本和源到目标映射必须显示你调查了以下内容。报告应仅总结材料决策：

- 主要对象、嵌套数组、重复的 XML 元素和特定源表示；
- 候选主键和外键；
- 每个源集合的粒度；
- 日期、时间戳、布尔值、货币、百分比和缺失值格式；
- 出现在一个或两个源中的字段；
- 源内的重复记录和跨源的重复记录；
- 转换前需要假设。

完成提供的源到目标映射模板。其目标字段已预填，但你必须调查并输入每个所需输出字段的适用 JSON 和/或 XML 路径、派生或规范化、重叠/冲突规则和笔记本证据。源路径故意未预填，因为分配的包中两个源布局不同。映射必须描述你的方法，而不是列出每个单个数据行

使用 JSON、XML、两者或派生完成六个空白列，用于源_格式；结构路径用于 json_源_路径和 xml_源_路径；以及转换_或_派生、重叠_或_冲突_规则和笔记本_证据。使用 | 分隔多个输入字段。解决方案笔记本模板解释了每一列，并包含虚构的示例。

任务 2：生成六个标准化的关系表

生成这六个提交的数据表：

1. 订单
2. 订单_商品
3. 客户
4. 配送
5. 产品
6. 产品_评价

使用这些文件名：

- GroupNNN_订单_标准化.csv
- GroupNNN_order_items_标准ised.csv
- GroupNNN_客户_标准化.csv
- GroupNNN_配送_标准化.csv
- GroupNNN_产品_标准化.csv
- GroupNNN_产品_评价_标准化.csv

所需的谷物和主键是：

Table	Grain	Primary key
orders	One row per canonical order	order_id
order_items	One row per order item	order_item_id
customers	One row per customer	customer_id
deliveries	One row per completed order delivery	delivery_id
products	One row per product	product_id
product_reviews	One row per canonical product review	review_id

Required foreign-key relationships are:

Child field	Parent field
orders.customer_id	customers.customer_id
order_items.order_id	orders.order_id
order_items.product_id	products.product_id
deliveries.order_id	orders.order_id
product_reviews.order_id	orders.order_id
product_reviews.order_item_id	order_items.order_item_id
product_reviews.product_id	products.product_id
product_reviews.customer_id	customers.customer_id

All listed primary and foreign keys are required and non-null.

For each file, follow `public_data_dictionary.csv` for table grain, field names, field order, data types, nullability and numeric comparison tolerance. Do not add helper or EDA-only columns to the six submitted CSV files. Helper tables and columns may be created within your code.

Your workflow must, where applicable:

- flatten nested arrays without changing the intended entity grain;
- retain valid one-to-many relationships rather than forcing all information into one flat table;

表	谷物	主键
订单	每个规范订单一行	订单ID_
按订单项排序_	每个订单项一行	订单项ID_ _
客户	每个客户一行	客户ID_
配送	每行一个完成的订单配送	配送_id
产品	每行一个产品	产品_ID
产品_评价	每个规范产品评价一行	评价ID_

必须的外键关系是：

子字段	父字段
订单.客户ID_	客户.客户ID_
订单项.订单ID_ _	订单.订单ID_
订单_项.产品_ID	产品产品_id
配送订单 _id	订单订单 _id
产品_评价订单_id	订单订单 id_
产品_评价订单_商品_id	订单项.订单项ID_ _ _
产品_评论.产品_ID	产品.产品_ID
产品_评论.客户_ID	客户.客户ID_

所有列出的主键和外键都是必需且不能为空。

对于每个文件，请遵循 `public_data_dictionary.csv` 以获取表格谷物、字段名称、字段顺序、数据类型、空值性和数值比较容差。不要向提交的六个 CSV 文件中添加辅助或仅用于 EDA 的列。辅助表和列可以在您的代码中创建。

您的流程必须，在适用的情况下：

- 展平嵌套数组而不改变预期的实体粒度；
- 保留有效的“一对多”关系，而不是将所有信息强制填入一个扁平化表格；

- normalise stable identifiers without inventing new entity identities;
- preserve identifier leading zeros and do not lower-case identifiers or structured categories unless a field-specific rule requires it;
- standardise dates as YYYY-MM-DD;
- standardise timestamps as YYYY-MM-DD HH:MM:SS;
- standardise booleans as True or False;
- remove currency labels and thousands separators before numeric conversion;
- represent percentages as the numeric percentage points required by the target field;
- derive fields required by the public dictionary, including line and order arithmetic where appropriate;
- use the literal string NaN for a prescribed missing string result rather than an empty string;
- reconcile records within and across sources using stable business keys and field-level comparison; and
- produce deterministic outputs when rerun on the same input.

For source reconciliation, normalise comparable fields first, compare records using the table primary key and retain one canonical row per key. Matching records in the assessed data are consistent after the published normalisation rules. A correctly normalised duplicate therefore does not require an arbitrary JSON-over-XML or XML-over-JSON precedence rule. If your code detects different non-missing values for the same target field and key, record the conflict in validation instead of silently choosing one.

For a prescribed missing string output, write the three literal characters NaN. This is not a Python None, NumPy/pandas missing value or empty field. When reading a submitted CSV for validation, consider pandas.read_csv(path, keep_default_na=False) so the sentinel remains visible. Do not place the string NaN in a required numeric or boolean field.

Use the following sequence for the published order arithmetic:

1. calculate each line_revenue as quantity * unit_price, rounded to two decimal places;
2. calculate order_price as the sum of the rounded line revenues, rounded to two decimal places;
3. treat product prices as GST-inclusive and calculate the included tax_amount as order_price / 11, rounded to two decimal places, before the coupon discount;
4. apply coupon_discount to order_price;
5. add delivery_charges; and
6. round the resulting order_total to two decimal places.

tax_amount is reported separately and is not added again to order_total. Use an absolute comparison tolerance of 0.01 for published monetary fields.

Do not hard-code canonical row counts, individual expected values or lists of known duplicate identifiers. A correct pipeline must derive its outputs from the allocated source files.

- 规范化稳定标识符，无需创建新的实体标识；
- 保留标识符的前导零，标识符或结构化类别不转换为小写，除非特定字段规则要求；
- 标准化日期为 YYYY-MM-DD；
- 标准化时间戳为 YYYY-MM-DD HH:MM:SS；
- 标准化布尔值为 True 或 False；
- 在数值转换前移除货币标签和千位分隔符；
- 以目标字段所需的数值百分比点表示百分比；
- 根据公共词典导出所需字段，并在适当情况下进行行和顺序算术运算；
- 对于规定的缺失字符串结果，使用字面量字符串 “NaN” 而不是空字符串；
- 使用稳定的业务键和字段级比较，在来源内部和跨来源之间协调记录；以及
- 在相同输入上重新运行时，产生确定性输出。

对于来源协调，首先规范化可比较字段，然后使用表主键比较记录，并保留每个键的一个规范行。在评估数据中匹配的记录在发布后的规范化规则下保持一致。因此，正确规范化的重复记录不需要任意的 JSON-over-XML 或 XML-over-JSON 优先级规则。如果您的代码检测到同一目标字段和键的不同非缺失值，请在验证中记录冲突，而不是静默地选择其中一个。

对于指定的缺失字符串输出，请写入三个字面字符 NaN。这不是 Python 的 None、NumPy/pandas 的缺失值或空字段。在读取提交的 CSV 文件进行验证时，请使用 pandas.read_csv(path, keep_default_na=False)，以便保留哨兵值。不要将字符串 NaN 放在必填的数字或布尔字段中。

对于发布的订单运算，请使用以下序列：

1. 计算_每行收入为数量 * 单价_，结果保留两位小数；
2. 计算_订单价格为所有四舍五入后的行收入之和，结果保留两位小数；
3. 将产品价格视为含 GST，并在优惠券折扣前计算含税金额_，即订单_价格 / 11，结果保留两位小数；
4. 对订单_价格应用优惠券_折扣；
5. 添加配送_费用；和
6. 将产生的订单_总额四舍五入到两位小数。

税_金额单独报告，不再添加到订单_总额中。对已发布的货币字段使用绝对比较容差 0.01。

不要硬编码规范行数、单个预期值或已知重复标识符的列表。正确的管道必须从分配的源文件中推导其输出。

Task 3: Regex and multilingual text preprocessing

Regular expressions are assessed for bounded narrative cleaning and extraction. They must be applied after the relevant JSON value or XML element has been obtained with a structured parser.

Raw narrative fields may contain HTML entities, HTML/XML-like tags, URLs, emoji, mixed whitespace, social markers and bracketed system metadata. For this assessment, the complete set of removable bracketed and social marker patterns is::

[SYSTEM]

[CATALOGUE]

[VERIFIED_PURCHASE]

[SOURCE: ...]

[RATING: n/5]

#verified-buyer

@store_support

Apply this text-processing order:

- 1. obtain the value using the structured JSON/XML parser;
- 2. extract any required order, SKU or promotion reference from the raw value;
- 3. decode HTML entities and apply Unicode NFC normalisation;
- 4. remove HTML/XML-like tags while retaining human-readable content;
- 5. remove the listed system/source/rating/social markers, URLs and emoji;
- 6. remove a complete review reference wrapper, including **Reference:**, the order reference, separator, **SKU:** and SKU;
- 7. remove **PROMO:** together with its code;
- 8. collapse spaces, tabs and line breaks, trim, and lower-case the designated cleaned narrative; and
- 9. return the literal string **NaN** if no human-readable text remains.

Use regex for bounded patterns such as tags, markers, URLs, whitespace and business-reference extraction. Standard-library tools such as **html** and **unicodedata** may be used for entity decoding, Unicode normalisation and script classification. Do not use regex to reconstruct JSON or XML structure.

Extract these embedded business references when present:

任务 3：正则表达式和多语言文本预处理

正则表达式用于有界叙事清理和提取，必须在使用结构化解析器获取相关 JSON 值或 XML 元素后应用。

原始叙事字段可能包含 HTML 实体、HTML/XML 样式的标签、URL、表情符号、混合空白、社交标记和括号内的系统元数据。对于本次评估，可移除的括号和社交标记模式完整集为：

[SYSTEM]

[目录]

[已验证_购买]

[来源： ...]

[评分： n/5]

#已验证买家

@店铺_支持

应用此文本处理顺序：

- 1. 使用结构化JSON/XML解析器获取值；
- 2. 从原始值中提取任何所需的订单、SKU或促销参考；
- 3. 解码 HTML 实体并应用 Unicode NFC 规范；
- 4. 移除 HTML/XML 类似标签，同时保留可读内容；
- 5. 移除列出的系统/来源/评分/社交标记、URL 和表情符号；
- 6. 移除完整的评论引用包装器，包括 **Reference:**、订单引用、分隔符、SKU: 和 SKU；
- 7. 移除 **PROMO:** 及其代码；
- 8. 合并空格、制表符和换行符，修剪，并将指定的清理后的叙述转换为小写；和
- 9. 如果没有人类可读的文本剩余，则返回字面量字符串 NaN。

使用正则表达式处理有界模式，例如标签、标记、URL、空白字符和业务参考提取。标准库工具如 **html** 和 **unicodedata** 可用于实体解码、Unicode 规范化和脚本分类。不要使用正则表达式来重建 JSON 或 XML 结构。

提取这些嵌入的业务参考（当存在时）：

Reference	Public format	Output when absent
Order reference	HORD or CORD followed by exactly six digits	NaN
Product SKU	SKU- followed by one or more ASCII letters or digits	NaN
Promotion code	B1SAVE- to B5SAVE-, followed by exactly two digits	NaN

Return extracted references in upper case. Boundary handling matters: a longer, malformed or embedded near-match must not be accepted as a valid reference.

Multilingual review contract

The reviews are primarily English but include a smaller multilingual component. The two review outputs have different purposes:

- review_body_clean must preserve valid multilingual UTF-8 letters. It must not erase a valid review merely because it contains non-Latin writing.
- review_body_latin_analysis is a separate derived field. It retains Latin-script letters, including European diacritics, together with applicable digits and punctuation, while removing non-Latin letters. If no Latin letter remains, return NaN.
- contains_non_latin_script is based on whether the cleaned multilingual review contains any letter outside the Latin script. A non-ASCII character is not automatically non-Latin.

Do not infer language_code from the Latin-analysis field. Use and reconcile the structured source evidence supplied for that attribute.

Build review_body_latin_analysis from review_body_clean, not from the noisy raw review. Derive the review measures as follows:

review_length_chars = number of Python characters in review_body_clean

review_word_count = number of whitespace-separated tokens in review_body_clean

When review_body_clean is the literal NaN, preserve the published sentinel behaviour rather than counting the three letters as an ordinary review.

Document and test your text functions with matched, unmatched, missing, multilingual and near-match examples. Run the supplied public text cases and add your own tests. Load the cases with keep_default_na=False when using pandas. The teaching team may test the submitted functions on private edge cases that follow this specification. Private tests will not require behaviour that is absent from the published rules.

参考	公共格式	缺失时输出
顺序参考	HORD或CORD后跟恰好六位数字	NaN
产品SKU	SKU-后跟一个或多个ASCII字母或数字	NaN
促销代码	B1SAVE-至B5SAVE- ，后跟恰好两个数字	NaN

以大写形式返回提取的参考。边界处理很重要：一个更长、格式错误或接近匹配嵌入的不能被视为有效参考。

多语言评审合同

评审内容主要为英语，但也包含较小的多语言部分。这两个评审输出有不同的用途：

- review_body_clean 必须保留有效的多语言 UTF-8 字母。它不能仅仅因为包含非拉丁文字就擦除一个有效的评审。
- review_body_latin_analysis 是一个独立的派生字段。它保留拉丁字母文字，包括欧洲变音符号，以及适用的数字和标点符号，同时移除非拉丁字母。如果没有拉丁字母剩余，返回 NaN。
- contains_non_latin_script 是基于清理后的多语言评审是否包含任何拉丁字母文字以外的字母。非 ASCII 字符不自动属于非拉丁文字。

不要从拉丁分析字段推断语言_代码。使用并协调为此属性提供的结构化源证据。

从 review_body_clean 构建 review_body_latin_分析，而不是从嘈杂的原始 review。按以下方式导出 review 指标：

review_长度_字符 = review_主体_干净

review_词_数 = review_正文_清理

当 review_body_clean 是字面 NaN 时，保留已发布的哨兵行为，而不是将三个字母计为普通的 review。

使用匹配、不匹配、缺失、多语言和近似匹配示例来记录和测试您的文本函数。运行提供的公共文本用例并添加您自己的测试。在使用 pandas 时，用 keep_default_na=False 加载用例。教学团队可能会根据本规范对提交的函数在私有边缘用例上进行测试。私有测试将不需要在已发布的规则中缺失的行为。

Place the following six functions in `GroupNNN_text_functions.py` so the published interface can be tested without editing your notebook:

```
def clean_narrative_text(value):
    """Accept None or a string; return cleaned text or the string 'NaN'."""

def extract_order_reference(value):
    """Accept None or a string; return the upper-case reference or 'NaN'."""

def extract_product_sku(value):
    """Accept None or a string; return the upper-case SKU or 'NaN'."""

def extract_promo_code(value):
    """Accept None or a string; return the upper-case code or 'NaN'."""

def build_latin_analysis(value):
    """Accept cleaned multilingual text; return Latin analysis or 'NaN'."""

def contains_non_latin_script(value):
    """Accept cleaned multilingual text; return a Python bool."""
```

The function names, argument count and return contract are fixed. Your implementation is not. Do not perform file I/O, network access or row-specific lookups inside these functions.

Task 4: Validate the processed data

Provide executable validation checks in the validation-register section of the solution notebook. Give each check a stable ID such as `VAL-PK-01` so that the report can cite it without copying the complete check evidence. For every check, show the observed result, PASS/FAIL status and resolution or interpretation. The notebook template also reminds you of the difference between the literal string `NaN`, an empty string and a Python/pandas missing value. At a minimum, cover:

- required table and column presence;
- target data types and missing-value representation;
- primary-key completeness and uniqueness;
- foreign-key integrity between the six tables;
- allowed categorical values and sensible numeric ranges;
- source coverage and row counts before and after major transformations;
- handling of within-source and cross-source overlap;
- order price reconciliation against order-item lines;
- order total reconciliation against price, discount and delivery fields, plus a separate check that tax is the included GST component and was not added again;
- valid temporal ordering among order, dispatch, promised, delivered and review timestamps where applicable;
- extracted order/SKU/promotion references; and
- multilingual preservation and non-Latin-script indicators.

将以下六个函数放在 `GroupNNN_text_functions.py` 中，以便在不修改您的笔记本的情况下测试已发布的接口：

```
def clean_narrative_text(value):
    """接受 None 或字符串；返回清理后的文本或字符串 'NaN'。"""

def extract_order_reference(value):
    """接受 None 或字符串；返回大写的参考或 'NaN'。"""

def extract_产品_sku(value):
    """接受 None 或字符串；返回大写的 SKU 或 'NaN'。"""

def extract_促销_码(value):
    """接受 None 或字符串；返回大写代码或 'NaN'。"""

def build_拉丁_分析(value):
    """接受清理后的多语言文本；返回拉丁分析或 'NaN'。"""

def contains_非_拉丁_文字(value):
    """接受清理后的多语言文本；返回一个 Python 布尔值。"""
```

函数名称、参数数量和返回契约是固定的。您的实现不是。在这些函数内部不要执行文件 I/O、网络访问或特定行查找。

任务 4：验证处理后的数据

在解决方案笔记本的验证注册部分提供可执行的验证检查。为每个检查分配一个稳定的 ID，例如 `VAL-PK-01`，以便报告可以引用它而无需复制完整的检查证据。对于每个检查，显示观察结果、PASS/FAIL 状态和解决方案或解释。笔记本模板还提醒您区分字面量字符串 `NaN`、空字符串和 Python/pandas 缺失值。至少应涵盖：

- 必需的表和列存在；
- 目标数据类型和缺失值表示；
- 主键完整性和唯一性；
- 六个表之间的外键完整性；
- 允许的分类值和合理的数值范围；
- 主要转换前后的源覆盖率与行数；
- 源内和跨源重叠的处理；
- 按订单项行对订单价格进行核对；
- 按价格、折扣和配送字段对订单总额进行核对，并单独检查税额是否为包含的 GST 部分且未被重复添加；
- 在适用情况下，订单、发货、承诺、交付和评审时间戳之间的有效时间顺序；
- 提取的订单/SKU/促销引用；以及
- 多语言保护和非拉丁脚本指示符。

Use clear tolerances for floating-point or currency checks. A validation result must show the check, observed result and pass/fail status. Do not write assertions that only confirm a hard-coded expected count.

Finding and reporting a genuine source or contract issue is professional data work. A reported failure is not automatically incorrect when the evidence and proposed treatment are justified. Do not fabricate a value solely to make an assertion pass.

Task 5: Conduct focused EDA

Conduct univariate, bivariate and multivariate analysis across the relational outputs. Automated profile dumps do not by themselves satisfy this task.

Your report must include 6-8 clearly labelled assessed visualisations. The set must contain at least one investigation from each of these six categories:

1. univariate distribution or composition;
2. bivariate relationship or group comparison;
3. multivariate or segmented relationship;
4. temporal pattern;
5. review/text behaviour, such as language, length, rating or helpfulness; and
6. delivery or operational performance in relation to an order, customer, product or review characteristic.

Collectively, the assessed visualisations must use evidence from at least four of the six output tables and include at least two analyses that correctly join related tables. One visualisation may satisfy more than one category when the analytical question genuinely does so. A coordinated set of subplots counts as one visualisation when the panels answer one shared question. Label the assessed items **Figure 1**, **Figure 2**, and so on. Only the first eight clearly labelled assessed visualisations are marked.

For every assessed visualisation:

1. state the analytical question;
2. state the observation unit and denominator;
3. identify the tables and join keys used;
4. choose a chart appropriate to the variables and comparison;
5. provide readable titles, axes, legends and units; and
6. explain the result and a material limitation.

Take particular care with joins. Joining several one-to-many tables can multiply rows and inflate revenue, order or customer metrics. Calculate each metric at its intended grain and demonstrate a check against accidental double counting.

对浮点数或货币检查使用明确的容差。验证结果必须显示检查、观察结果和通过/失败状态。不要编写仅确认硬编码预期计数的断言。

发现并报告真实的来源或合同问题是一项专业的数据工作。当证据和拟议的处理方案具有充分理由时，报告的失败情况并不自动意味着错误。不要编造一个值，仅仅为了让一个断言成立。

任务 5：执行聚焦式 EDA

对关系输出进行单变量、双变量和多变量分析。自动生成的配置转储本身并不能满足这项任务。

你的报告必须包含 6-8 个清晰标注的评估可视化图表。该集合必须至少包含来自以下六个类别中的每个类别的一项调查：

1. 单变量分布或组成；
2. 双变量关系或分组比较；
3. 多元或分段关系；
4. 时间模式；
5. 审核或文本行为，例如语言、长度、评分或有用性；和
6. 与订单、客户、产品或评论相关的交付或运营绩效特征。

总体而言，评估的可视化必须使用至少六张输出表中的四张证据，并包含至少两项正确连接相关表的分析。当分析问题确实如此时，一个可视化可能满足多个类别。一套协调的子图在面板回答一个共同问题时算作一个可视化。将评估的项目标记为图1、图2等。只有前八个清晰标记的评估可视化会被评分。

对于每个评估的可视化：

1. 说明分析问题；
2. 说明观测单位和分母；
3. 确定使用的表格和连接键；
4. 选择适合变量和比较的图表；
5. 提供可读的标题、坐标轴、图例和单位；以及
6. 解释结果和材料限制。

在处理连接时要特别小心。连接多个一对多表可能会使行数成倍增加，并导致收入、订单或客户指标虚高。应在其预期粒度上计算每个指标，并展示防止意外重复计算的检查。

Task 6: Present findings and future ML questions

Present exactly ten numbered, evidence-based findings. Each finding must cite an assessed figure or reported statistic and explain:

- what was observed and at what grain;
- the magnitude, denominator or sample size;
- why it may matter in the retail context;
- a plausible alternative explanation or uncertainty; and
- a proportionate implication, next investigation or recommendation.

A finding that an apparent relationship is weak, uncertain or not decision relevant can be high quality when supported by evidence. Do not claim causation from observational association without a defensible design.

Propose exactly five numbered machine-learning questions grounded in your EDA. Together they must cover at least two different problem types, such as classification, regression, clustering, anomaly detection or forecasting. For each question, specify:

- the business decision or use case;
- the prediction or analysis unit;
- the proposed target or unsupervised objective;
- candidate predictors available at the decision time;
- an appropriate validation split;
- at least one evaluation metric; and
- likely target leakage, temporal leakage, fairness or deployment risks.

The questions must be feasible with the available type of data. Use the supplied EDA notebook template and aim for approximately 80-120 words or one compact table per question. Do not train or compare models in A1.

5. Reproducibility and code requirements

Submit Python code that recreates all six CSV files and validation results from the allocated raw package, plus an EDA notebook that recreates all assessed visualisations and reported statistics from those six CSVs.

Both submitted notebook must:

- contain a clearly labelled configuration cell; the solution notebook must define `GROUP_ID`, `INPUT_DIR` and `OUTPUT_DIR`;
- use relative or configurable paths rather than a student's absolute path;
- run from a fresh kernel using Restart and Run All;
- include the output required to review the analysis;
- avoid manual editing of generated CSV files;
- avoid hard-coded canonical answers; and
- use vectorised or otherwise reasonable methods for the supplied scale.

任务6：展示发现和未来的机器学习问题

展示正好十个编号的、有证据支持的发现。每个发现必须引用一个评估的数值或报告的统计数据，并解释：

- 观察到了什么以及分析的粒度；
- 幅度、分母或样本量；
- 为什么这在零售环境中可能很重要；
- 一个合理的替代解释或不确定性；以及
- 一个相称的影响、下一步的调查或建议。

一项发现表明，若一个明显的关系较弱、不确定或不具有决策相关性，并且有证据支持，则该发现仍可能是高质量的。在没有可辩护的设计的情况下，不要从观察性关联中声称因果关系。

请提出五个编号的机器学习问题，这些问题需基于你的探索性数据分析（EDA）。这五个问题必须涵盖至少两种不同的问题类型，例如分类、回归、聚类、异常检测或预测。对于每个问题，请指定：

- 相关的业务决策或用例；
- 预测或分析单元；
- 建议的目标或无监督目标。
- 决策时可供使用的候选预测变量；
- 适当的验证拆分；
- 至少一个评估指标；以及
- 可能存在目标泄漏、时间序列泄漏、公平性或部署风险。

问题必须在使用的数据类型上可行。使用提供的EDA笔记本模板，每道问题目标为80-120字或一个紧凑表格。不要在A1中训练或比较模型。

5. 可重复性和代码要求

提交 Python 代码，用于重现分配的原始包中的所有六个 CSV 文件和验证结果，并附带一个 EDA 笔记本，用于重现从这六个 CSV 文件中评估的所有可视化结果和报告的统计数据。

提交的笔记本必须：

- 包含一个清晰标记的配置单元格；解决方案笔记本必须定义 `GROUP_ID`、`INPUT_DIR` 和 `OUTPUT_DIR`；
- 使用相对路径或可配置路径，而不是学生的绝对路径；
- 使用 Restart and Run All 从一个全新的内核运行；
- 包含用于审查分析所需的输出；
- 避免手动编辑生成的 CSV 文件；
- 避免使用硬编码的规范答案；并且
- 使用向量化或其他合理的方法来处理提供的规模。

The solution notebook must contain the complete transformation and validation workflow. The EDA notebook must load the submitted six CSVs and recreate the reported figures and statistics without rerunning a different hidden cleaning pipeline. The submitted `GroupNNN_solution.py` export must reflect the solution notebook. It may import the required functions from `GroupNNN_text_functions.py`; the functions do not need to be duplicated. List any non-standard dependencies and versions in `requirements.txt`. The assessed workflow must run offline: it must not depend on a live AI service, external API, network download, private account, API key or unsubmitted local file.

Markers are not required to repair paths, install undocumented packages, rewrite student code or manually complete a failed transformation. Where a notebook cannot execute, available output and documented method may still provide partial evidence under the rubric.

6. EDA report requirements

Submit `GroupNNN_EDA.pdf`, with a maximum of 10 pages from the introduction to the conclusion. The cover page, contents page and references are excluded from the limit. Appendices are not assessed unless the specification explicitly requires the material or the main report directly cites it as validation evidence.

Use a clear structure such as:

1. problem context and data scope;
2. data-preparation assurance;
3. 6-8 assessed visualisations;
4. 10 numbered findings;
5. 5 numbered ML questions;
6. limitations; and
7. conclusion.

The data-preparation assurance section should communicate approximately 3-5 material transformation decisions and 4-6 material validation results, citing stable mapping and validation IDs where useful. The separate mapping CSV remains the complete field-lineage record, and the solution notebook remains the complete executable check record. Do not repeat either full artifact in the report.

Code listings do not belong in the report unless a short expression is necessary to explain an assessed decision. Tables and figures must remain readable at 100% zoom. Cite external facts, code, data and ideas using a consistent referencing style.

7. Submission requirements

Submit these two files to the A1 Moodle assignment:

- `GroupNNN_A1_submission.zip`

解决方案笔记本必须包含完整的转换和验证工作流程。EDA 笔记本必须加载提交的六个 CSV 文件，并重现报告的图表和统计数据，而无需重新运行不同的隐藏清洗管道。提交的 `GroupNNN_solution.py` 导出必须反映解决方案笔记本。可以导入 `GroupNNN_text_functions.py` 中的所需函数；函数无需重复。在 `requirements.txt` 中列出任何非标准依赖项和版本。评估的工作流程必须在离线状态下运行：它不能依赖于实时 AI 服务、外部 API、网络下载、私有账户、API 密钥或未提交的本地文件。

修复路径、安装未记录的软件包、重写学生代码或手动完成失败的转换，都不需要标记。当笔记本无法执行时，可用的输出和记录的方法可能根据评分标准提供部分证据。

6. EDA报告要求

提交 `GroupNNN_EDA.pdf`，最多10页，从引言到结论。封面、目录和参考文献不计入页数限制。附录只有在规格明确要求材料或主报告直接引用其作为验证证据时才会被评估。

使用清晰的格式，例如：

1. 问题背景和数据范围；
2. 数据准备保证；
3. 6-8个评估可视化；
4. 10个编号发现；
5. 5个编号的机器学习问题；
6. 限制；以及
7. 结论。

数据准备保证部分应传达大约3-5个实质性转换决策和4-6个实质性验证结果，在有用时引用稳定的映射和验证ID。单独的映射CSV文件是完整的字段血缘记录，解决方案笔记本是完整的可执行检查记录。报告中不要重复任何完整工件。

代码列表不应出现在报告中，除非需要简短的表达来解释一个评估的决策。表格和图表必须在100%缩放下保持可读性。使用一致的引用风格来引用外部事实、代码、数据和想法。

7. 提交要求

将这两个文件提交到A1 Moodle作业：

- `GroupNNN_A1_提交.zip`

- GroupNNN_EDA.pdf

The ZIP file must contain:

- GroupNNN_solution.ipynb
- GroupNNN_solution.py
- GroupNNN_EDA.ipynb
- GroupNNN_text_functions.py
- GroupNNN_source_to_target_mapping.csv
- GroupNNN_AI_declaration.pdf
- requirements.txt # required only for non-standard dependencies
- AI_records/ # required when conversational AI was used
 - GroupNNN_AI_index.pdf
 - complete_chat_exports...
- outputs/
 - GroupNNN_orders_standardised.csv
 - GroupNNN_order_items_standardised.csv
 - GroupNNN_customers_standardised.csv
 - GroupNNN_deliveries_standardised.csv
 - GroupNNN_products_standardised.csv
 - GroupNNN_product_reviews_standardised.csv

Do not submit the raw JSON/XML files supplied by the teaching team. Do not place the EDA PDF inside the ZIP. Follow the checklist in Appendix A before uploading. Every group member is responsible for verifying that the correct group files appear in the Moodle submission receipt.

8. Generative AI use

Generative AI tools may be used while preparing A1, subject to University and unit policy. The group remains responsible for every submitted transformation, test, figure, citation, statement and file.

Every group must submit the signed AI-use declaration. If any member used a conversational AI tool, the group must also submit a complete direct export of every assignment-related conversation in [AI_records/](#), together with an English index that records purpose, submission location and independent verification. Do not selectively shorten an assignment conversation. Use separate assessment-specific chats so unrelated private material is not mixed into the record. If a tool has no export function, print the complete conversation to PDF or HTML. Inline code completion with no conversational history must still be declared, but does not require an invented chat export. Follow [templates/A1_AI_use_declaration_template.docx](#).

- GroupNNN_EDA.pdf

ZIP 文件必须包含：

- GroupNNN_解决方案.ipynb
- GroupNNN_solution.py
- GroupNNN_EDA.ipynb
- GroupNNN_文本_函数.py
- GroupNNN_source_to_target_映射.csv
- GroupNNN_AI_声明.pdf
- requirements.txt # 仅用于非标准依赖
- AI_记录/ # 当使用对话式AI时需要
 - GroupNNN_AI_索引.pdf
 - 完成_chat_导出...
- outputs/
 - GroupNNN_订单_标准化.csv
 - GroupNNN_订单_商品_标准化.csv
 - GroupNNN_客户_标准化.csv
 - GroupNNN_配送_标准化.csv
 - GroupNNN_产品_标准化.csv
 - GroupNNN_产品_评价_标准化.csv

不要提交教学团队提供的原始 JSON/XML 文件。不要将 EDA PDF 放在 ZIP 压缩包内。上传前请遵循附录 A 中的清单。每个小组成员都有责任确认 Moodle 提交收据中显示的是正确的组文件。

8. 生成式 AI 使用

在准备 A1 时，可以使用生成式 AI 工具，但需遵守大学和课程的政策。小组对所有提交的转换、测试、图表、引用、陈述和文件都负有责任。

每个小组都必须提交签署的AI使用声明。如果任何成员使用了对话式AI工具，该小组还必须提交AI_records/中所有与作业相关的对话的完整直接导出，并附带一个记录目的、提交位置和独立验证的英文索引。不要选择性地缩短作业对话。使用单独的评估特定聊天，以便无关的私人材料不会被混入记录中。如果工具没有导出功能，将完整对话打印为PDF或HTML。没有对话历史的内联代码补全仍需声明，但不需要虚构的聊天导出。遵循模板/[A1_AI_use_declaration_template.docx](#)。

Appendix A: Submission checklist

Use this checklist after restarting and running the final notebook.

Group and package

- ☐ The allocated archive, manifest and raw filenames show the same **GroupNNN** identifier.
- ☐ Every submitted filename uses that exact three-digit identifier.
- ☐ Every group member has reviewed the final Moodle submission receipt.

Reproducibility

- ☐ **GroupNNN_solution.ipynb** and **GroupNNN_EDA.ipynb** both run using Restart and Run All.
- ☐ The notebook has configurable **GROUP_ID**, **INPUT_DIR** and **OUTPUT_DIR** values and no student-specific absolute paths.
- ☐ **GroupNNN_solution.py** reflects the final notebook workflow.
- ☐ **GroupNNN_text_functions.py** exposes all six required functions and performs no file I/O or network access.
- ☐ The solution workflow recreates the CSVs and checks; the EDA workflow recreates figures and reported statistics without manual edits or live services.
- ☐ **requirements.txt** is included if non-standard dependencies are required.

Data products and assurance

- ☐ All six required CSVs are inside **outputs/**.
- ☐ Filenames, columns, order, data types and grains match the public data dictionary.
- ☐ Primary keys, foreign keys, source coverage, arithmetic and time are checked.
- ☐ GST, discount, delivery and total follow the published calculation order.
- ☐ Duplicate reconciliation is data-driven rather than hard-coded.
- ☐ The mapping completes every pre-filled target row using stable **MAP-...** IDs.
- ☐ The solution notebook records stable **VAL-...** IDs, observed results, status, evidence and interpretation.

Text processing

- ☐ JSON and XML were parsed structurally before regex was applied to text.
- ☐ Tags, entities, markers, URLs, social tokens, emoji and whitespace follow the published rules.
- ☐ Order, SKU and promotion references have boundary and near-match tests.
- ☐ **review_body_clean** preserves valid multilingual text.
- ☐ The separate Latin-analysis field preserves Latin diacritics.
- ☐ Every supplied public text case was run and additional tests are shown.

附录 A：提交清单

重启并运行最终笔记本后，使用此清单。

分组和打包

- ☐ 分配的存档、清单和原始文件名都显示相同的 GroupNNN 标识符。
- ☐ 每个提交的文件名都使用该确切的三个数字标识符。
- ☐ 每个小组成员都已审查最终的 Moodle 提交收据。

可重复性

- ☐ GroupNNN_solution.ipynb 和 GroupNNN_EDA.ipynb 都使用 Restart and Run All 来运行。
- ☐ 该笔记本具有可配置的 GROUP_ID、INPUT_DIR 和 OUTPUT_DIR 值，并且没有学生特定的绝对路径。
- ☐ GroupNNN_solution.py 反映了最终的笔记本工作流程。
- ☐ GroupNNN_text_functions.py 暴露了所有六个必需的函数，并且不执行文件 I/O 或网络访问。
- ☐ 解决方案工作流程重新创建了 CSV 文件和检查；EDA 工作流程重新创建了图表和报告的统计数据，无需手动编辑或实时服务。
- ☐ 如果需要非标准依赖项，则会包含 requirements.txt。

数据产品与保证

- ☐ 所有六个必需的 CSV 文件都在 outputs/ 目录内。
- ☐ 文件名、列、顺序、数据类型和粒度与公共数据字典一致。
- ☐ 主键、外键、源覆盖率、计算和时效性均经过检查。
- ☐ GST、折扣、配送和总计遵循已发布的计算顺序。
- ☐ 重复对账采用数据驱动而非硬编码的方式。
- ☐ 该映射使用稳定的 MAP-... ID 完成每个预填充的目标行。
- ☐ 解决方案笔记本记录稳定的 VAL-... ID，观察到的结果，status，证据和解释。

文本处理

- ☐ JSON 和 XML 在应用正则表达式到文本之前被结构化解析。
- ☐ 标签、实体、标记、URL、社交令牌、表情符号和空白符遵循已发布的规则。
- ☐ 订单、SKU 和促销引用有边界和近似匹配测试。
- ☐ review_body_clean 保留有效的多语言文本。
- ☐ 单独的拉丁分析字段保留了拉丁变音符号。
- ☐ 所有提供的公共文本案例均已运行，并展示了附加测试。

Report

- ☐ GroupNNN_EDA.pdf is no more than 10 assessed pages.
- ☐ Between 6 and 8 figures are labelled as assessed visualisations.
- ☐ The set covers all six analytical categories, at least four tables and at least two correct relational analyses.
- ☐ Observation units, denominators and join-multiplication checks are clear.
- ☐ Exactly 10 evidence-based findings and 5 ML questions are numbered.
- ☐ Limitations, uncertainty, leakage risks and external citations are clear.

AI and final archive

- ☐ GroupNNN_AI_declaration.pdf is signed by all members.
- ☐ When conversational AI was used, AI_records/ contains the index and complete assignment-related exports.
- ☐ Inline completion is declared even when no chat export exists.
- ☐ Material AI-generated code and claims have independent checks.
- ☐ Raw teaching data is not included in the submission.
- ☐ The final uploaded files are exactly GroupNNN_A1_submission.zip and GroupNNN_EDA.pdf.

Appendix B: Glossary

Term	Meaning in A1
Canonical row	The single retained row representing one business entity after overlap is reconciled
Data grain	What one row represents, for example one order or one order item
Denominator	The total population used to calculate a rate or proportion
Deterministic	The same input and code produce the same output every time
Foreign key	A field that refers to a primary key in another table
Lineage	Where an output field came from and how it was transformed

报告

- ☐ GroupNNN_EDA.pdf 最多不超过 10 页评估内容。
- ☐ 其中 6 至 8 个图表被标记为评估过的可视化内容。
- ☐ 该集合涵盖了所有六个分析类别，至少包含四个表格，以及至少两个正确的关联分析。
- ☐ 观察单元、分母和连接乘法检查都很清晰。
- ☐ 正好有 10 个基于证据的发现和 5 个机器学习问题被编号。
- ☐ 局限性、不确定性、泄漏风险和外部引用都很明确。

人工智能与最终存档

- ☐ GroupNNN_人工智能_声明.pdf 已由所有成员签署。
- ☐ 当使用对话式人工智能时，AI_记录/ 包含索引和完整的任务相关导出。
- ☐ 即使没有聊天导出，也会声明内联完成。
- ☐ 材料人工智能生成的代码和声明具有独立的检查。
- ☐ 原始教学数据未包含在提交中。
- ☐ 最终上传的文件正好是 GroupNNN_A1_submission.zip 和 GroupNNN_EDA.pdf。

附录 B：术语表

Term	A1 中的含义
规范行	在重叠后，代表一个业务实体的单个保留行经过对账
数据粒度	一行代表什么，例如一个订单或一个订单项
分母	用于计算比率或比例的总人口或比例
确定性	相同的输入和代码每次都会产生相同的输出
外键	一个字段，它引用另一个表中的主键
血缘关系	一个输出字段来自哪里以及它是如何转换的

Term	Meaning in A1
Near-match	Text resembling a valid reference but failing a required boundary
Observation unit	The entity measured in an analysis, such as an order or review
Primary key	Field or fields that uniquely identify one row at the table grain
Relational join	Combining tables through related key fields
Source-to-target mapping	A record of each output field's source and transformation
Target leakage	Using information unavailable when a prediction must be made
Temporal leakage	Using future information to predict an earlier event
Unicode	A standard representing writing systems and symbols across languages
Vectorised operation	Applying an operation to a column/array rather than looping over every row

Term	A1中的含义
近似匹配	看似有效引用但未满足必要边界的文本
观测单元	分析中测量的实体，例如订单或评论
主键	用于唯一标识表中每一行的字段或字段在表粒度上
关系连接	通过关键字段组合表
源到目标映射	每个输出字段的源记录和转换
目标泄漏	使用在需要做出预测时不可用的信息
时间泄漏	使用未来信息来预测一个较早的事件
Unicode	一种代表不同语言书写系统和符号的标准
矢量化操作	将操作应用于列/数组，而不是逐行循环